Translated from Italian to English - www.onlinedoctranslator.com

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

SPECIFIC
TECHNICAL

Gas measurement systems

Hourly gas measuring devices

Part 11-3: Local Interface Communication Profile

UNI/TS 11291-11-3

MARCH 2014

Gas measurement systems

Hourly based gas metering systems

Part 11-3: Communication profile of local interface

The technical specification complements the technical regulatory requirements defined in the applicable sections of the UNI/TS 11291 technical specifications, enabling the interchangeability of natural gas metering units intended for the mass market (<G10) as well as other equipment operating in point-to-multipoint networks. The technical specification addresses the local interface communication profile.

ITALIAN TEXT

ICS 91.140.40

UNI
Italian National Unification Body
Via Sannio, 2
20137 Milan, Italy

© UNI
Reproduction prohibited. All rights reserved. No part of this document may be reproduced or distributed by any means, photocopying, microfilm, or otherwise, without the written consent of UNI.

www.uni.com

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

PREMISE

The technical specification is reviewed every three years.
Any comments on UNI/TS 11291-11-3 must reach UNI by March 2016.

This technical specification was developed under the jurisdiction of the body federated to UNI

CIG - Italian Gas Committee

The UNI Central Technical Commission gave its approval on 19 February 2014.

This technical specification was ratified by the President of UNI and became part of the national regulatory body on 13 March 2014.

UNI standards are developed by taking into account the views of all interested parties and reconciling any conflicting aspects, to represent the true state of the art on the subject and the necessary degree of consensus.
Anyone who, following the application of this standard, feels they can provide suggestions for its improvement or for its adaptation to the evolving state of the art is asked to send their contributions to UNI, the Italian National Unification Body, which will take them into consideration for any revision of the standard itself.

UNI standards are revised, when necessary, with the publication of new editions or updates.
It is therefore important that users ensure they have the latest edition and any updates.
Users are also invited to check the existence of UNI standards corresponding to the EN or ISO standards where cited in the regulatory references.

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

INDEX

0		**INTRODUCTION**	1
1		**PURPOSE AND SCOPE**	1
2		**REGULATORY REFERENCES**	2
3		**TERMS AND DEFINITIONS, ABBREVIATIONS**	3
3.1		Terms and definitions	3
3.2		Abbreviations	3
	prospectus 1	Abbreviations	3
4		**GENERAL INFORMATION**	3
	figure 1	Reference architecture	4
5		**LOCAL COMMUNICATION ARCHITECTURE**	4
5.1		Physical level	4
5.2		Data Link Layer	5
	prospectus 2	Clarifications on supported HDLC services for interchangeability	5
5.3		Application Layer	6
	prospectus 3	Clients allowed local access to GdMs/Repeaters	6
	prospectus 4	Clients that are allowed local access to the DCs/GWs	6
6		**WORKFLOWS**	7
6.1		Pre-protected commands	7
	figure 2	Sequence diagram showing interactions between different subjects, with commands pre-protected	8
6.2		Safe Commands on the HHT	9
6.3		Installation procedure	9
	figure 3	UNI/TS state diagram and RF state for PM1 devices (GdM and DC/GW/Repeaters)	10
	prospectus 5	Objects to be programmed that must be present at the end of the process installation	10
	prospectus 6	Objects to be programmed that must be present at the end of the process installation	11
6.4		Local actions	11
6.5		Firmware Update	11
	figure 4	Local firmware update	12
	figure 5	Firmware update items	13
	prospectus 7	Firmware update steps	13
6.6		Management of manufacturer-specific elements	13
APPENDIX TO (regulations)		**EXCHANGE FORMAT FOR PRE-PROTECTED COMMANDS AND RESPONSES**	14
	prospectus A.1		14
	prospectus A.2		15
APPENDIX B (regulations)		**HDLC FRAME STRUCTURE**	16
B.1		HDLC textures	16
	prospectus B.1		17
		BIBLIOGRAPHY	19

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

0 INTRODUCTION

On October 22, 2008, the Italian Electricity and Gas Authority published Resolution ARG/gas 155/08, "Directive for the commissioning of gas metering units, characterized by minimum functional requirements and with remote reading or remote management functions, for delivery points of natural gas distribution networks." This resolution was repealed by Resolution 631/2013/R/GAS of December 27, 2013, as it was incorporated into Resolutions 575/2012/R/gas and 573/2013/R/gas. Amendments to the text of this technical specification may become necessary as a result of measures adopted by the AEEG following this resolution.

The UNI/TS 11291-11 technical specification is composed of the following 6 parts, the full application of which is necessary for interchangeability purposes[1]:

- UNI/TS 11291-11-1 Gas measurement systems - Hourly gas measurement devices - Part 11-1: General
- UNI/TS 11291-11-2 Gas measurement systems - Hourly gas measurement devices - Part 11-2: Data model
- UNI/TS 11291-11-3 Gas measurement systems - Hourly gas measurement devices - Part 11-3: Local interface communication profile
- UNI/TS 11291-11-4 Gas measurement systems - Hourly gas measurement devices - Part 11-4: PM1 communication profile
- UNI/TS 11291-11-5 Gas measurement systems - Hourly gas measurement devices - Part 11-5: PP3 communication profile
- UNI/TS 11291-11-6 Gas measurement systems - Hourly gas measurement devices - Part 11-6: Specifications for conformity assessment

Part 11-1 includes the overall architecture definition and the subset of use cases specified for interchangeability.

Part 11-2 defines the subset of the data model specified for interchangeability.

Parts 11-3, 11-4 and 11-5 define the communication profile for each of the interchangeable interfaces: local ports, PM1 and PP3.

Part 11-6 defines a common specification framework for conformity assessment.

1 PURPOSE AND SCOPE

This technical specification completes the technical regulatory requirements, defined in the applicable parts of the UNI/TS 11291 specifications, in order to allow the interchangeability of natural gas measuring groups intended for the “mass market” (<G10) as well as other equipment operating in point-to-multipoint networks.

This technical specification applies to GdMs serving the Delivery Points of distribution systems with a maximum capacity of less than 10 m^3/h at reference conditions (or counter <G10), which communicate with point-to-multipoint protocol.

It also applies to network devices: DC/GW, translator, repeater.

This technical specification defines the communication profile of local interfaces that must be made available by measuring groups and network devices for installation and maintenance purposes.

1) A technical report will include specifications for possible upgrades and additional services, which however are not part of the minimum mandatory set for interchangeability.

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

Additionally, local maintenance devices (HHTs), although intended to be interchangeable, are not based on the same hardware and are often supplied independently by distribution companies. For this reason, this document specifies standard interfaces on the local port to support local interventions interchangeably. However, it should be noted that specifying the characteristics of HHTs is outside the scope of this technical specification.

This technical specification is to be used in conjunction with the relevant parts of the UNI/TS 11291 series (parts 1, 6, 7, 8 and 10) and the applicable international standards.

Note For reasons of interchangeability, this technical specification includes detailed interpretations of the standards and specific use cases, objects, and data structures.
These elements are mandatory for interchangeability purposes, however care has been taken not to prevent manufacturers from providing extensions and alternative mechanisms permitted by the protocols: however, these are not part of this document.

2 REGULATORY REFERENCES

This technical specification refers, through dated and undated references, to provisions contained in other publications. These normative references are cited at the appropriate points in the text and are listed below. For dated references, subsequent amendments or revisions made to those publications are valid only if introduced into this technical specification as an update or revision. For undated references, the latest edition of the referenced publication (including updates) applies.

UNI/TS 11291-1:2013	Gas measurement systems - Hourly gas measurement devices - Part 1: General characteristics of the remote management or remote reading system
UNI/TS 11291-6:2013	Gas measurement systems - Hourly gas measurement devices - Part 6: Requirements for measuring sets with ranges less than 10 m^3/h (counter <G10)
UNI/TS 11291-10	Gas measurement systems - Hourly gas measurement devices - Part 10: Safety
UNI/TS 11291-11-2:2014	Gas measurement systems - Hourly gas measurement devices - Part 11-2: Data model
IEC EN 62056-21	Electricity metering - Data exchange for meter reading, tariff and load control - Part 21: Direct local data exchange
ISO/IEC 13239	Information technology - Telecommunications and information exchange between systems - High-level data link control (HDLC) procedures
ISO 1177	Information processing - Character structure for start/stop and synchronous character oriented transmission
IEC 62056-5-3	Electricity metering data exchange - The DLMS/COSEM suite - Part 5-3: DLMS/COSEM application layer
EN 62056-6-1	Electricity metering data exchange - The DLMS/COSEM suite - Part 6-1: Object Identification System (OBIS)
EN 62056-6-2	Electricity metering data exchange - The DLMS/COSEM suite - Part 6-2: COSEM interface classes

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

3 TERMS AND DEFINITIONS, ABBREVIATIONS

3.1 Terms and definitions[2)]

For the purposes of this standard, the terms and definitions given in IEC 62056-5-3, EN 62056-6-1 and EN 62056-6-2 apply.

3.2 Abbreviations

For the purposes of this document, the following abbreviations apply.

prospectus 1 **Abbreviations**

Abbreviation	Meaning
APDU	Application Protocol Data Unit
SAC	Central access system as defined in point 3.1 of UNI/TS 11291-1:2013
CIU	User display - Customer information unit
GPRS	General Packet Radio Service
GSM	Global System for Mobile communications
HHT, HHU	Hand held terminal or Unit, local user terminal
ID	Identifier
MU, GdM	Measuring Group
SAP	Service Access Point
UC	Use Case / Use Case
RTC	Real Time Clock / Real Time Clock
LLS	Low-Level Security
HLS	High Level Security
TOU	Time Of Use / Time-slot management
UML	Unified Modeling Language
XML	Extensible Markup Language
AA	Application Association
HDLC	High-level Data Link Control
OBIS	Object Identification System - Object Identification System
DST	Daylight Saving Time - Daylight saving time
AARQ	A-Associate Request
AARE	A-Associate Response
RLRQ	A-Release Request
RLRE	A-Release Response

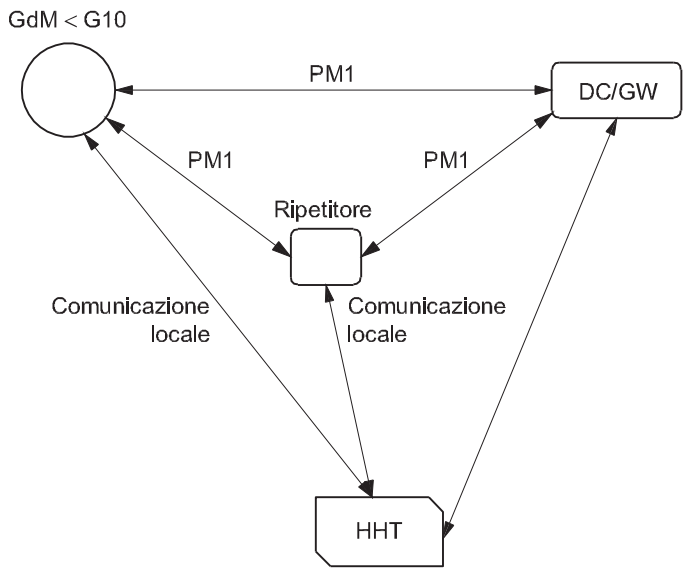
4 GENERAL INFORMATION

This document provides instructions on how to access the device – GdM, DC, GW or repeater – through the local interface.

It defines the physical layer, the data link layer and the application to be used, together with workflows that show the different interactions (figure 1).

2) See also COSEM Blue book, Green book, Yellow book, White book.

figure 1 **Reference architecture**



The local interface shall conform, as applicable, to:

- DLMS Users Association 1000-1 “Blue book” ed. 11 (hereinafter referred to as Blue Book);

- DLMS Users Association 1000-2 “Green Book” ed. 7 and related Amendment 3 (hereinafter referred to as the Green Book);

- Pre-established associations (point 9.4.4.4 of the Green Book); with

the specific extensions:

- Client user identification (Association LN Class_id: 15, version 2)

- Compact data (forthcoming).

5 LOCAL COMMUNICATION ARCHITECTURE

5.1 Physical level

5.1.1 Interface type

It must comply with IEC EN 62056-21.

5.1.2 Construction

It must comply with IEC EN 62056-21. Connection via optical reading head from the front panel (automatic positioning / magnetic fixing).

5.1.3 Optical characteristics

They must comply with IEC EN 62056-21.

5.1.4 Transmission

The optical interface must comply with IEC EN 62056-21; serial data transmission, asynchronous data according to ISO 1177, half duplex.

- Speed: 9 600 Bd

- Format: 1 start, 8 data bits, 0 parity (no parity) and 1 stop bit (9 600 8N1).

With reference to CEI EN 62056-21, the system must switch directly to HDLC mode completely skipping the request and preparation for the physical connection (i.e. HDLC *from the beginning*as also admitted by point 6.1 of the Green Book).

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

5.2 Data Link Layer

It must comply with:

- CEI EN 62056-21;
- ISO/IEC 13239;
- Green Book, point 8.

5.2.1 Plot structure

It must comply with the following standards:

- CEI EN 62056-21;
- ISO/IEC 13239;
- Green Book, point 8.4.

Further details are provided in Appendix B, which also provides some examples.

The client must comply with the requirements of Section 8 of the Green Book. For the purposes of interchangeability, some limitations may apply to the services provided, as detailed below. However, this does not limit manufacturers' ability to provide the full range of HDLC services as specified in the Green Book.

Table 2 refers to the points in the Green Book and provides the relevant clarifications. **Clarifications on supported HDLC services for interchangeability**

prospectus 2

Reference to Green Book requirement1)	Clarification
8.4.1.2 Flag field	The "continuously transmitted" frame mode is not used. This is achieved by setting the window size to 1.
8.4.1.3 Frame format field	The maximum frame length is defined to match the maximum size of the APDUs traveling over PM1, as the same application frames are expected to be used for interchangeability; however, the HHU must still explicitly send its transmit and receive length information field to the server so that both stations can agree on it.
8.4.2 MAC addressing	Since HDLC is used for local communication with an optical port, the MAC address on the server side (the end device) must use an address byte. This address must be the logical device address. Note MAC addressing as defined in the DLMS Green Book can use 1, 2 or 4 bytes on the server side while, on the client side it is always one byte: the client ID.
8.4.2.3 Reserved special HDLC addresses	No additional addresses are required other than those currently defined for the Italian specification.
8.4.2.4 Handling special addresses	Only the format defined for the Italian specification must be managed.
8.4.3.1 Selected repertoire and control field format	This technical specification does not support (from both sides: end device, HHU): RNR, UI.
8.4.3.5 Set normal response mode (SNRM) command	There is a risk of issuing simultaneous commands via PM1 and the local interface, which can generate programming inconsistencies. Therefore, as long as the local session is active, no programming commands/ requests on PM1 are accepted. Note The push function on the PM1 is not affected and may still work in any case.
8.4.5.4.2 Exchange of information frames	All frames sent by the client must have the “poll bit” set (1); all frames sent by the server must have the “final bit” set (1). The method in example 2 of the above point must be used.
8.4.5.4.3 Window size considerations	Window size for this specification = 1 (non-negotiable for interchangeability).
8.4.5.4.5 Transferring long MSDUs from the server to the client	“Long MSDUs” are not maintained for interchangeability purposes.
8.4.5.4.6 Multi- and broadcasting	Not managed for interchangeability purposes.
8.4.5.4.7 Sending an UI frame from the server to the client(push)	Not managed for interchangeability purposes.
8.4.5.4.8 Handling the CALLING device physical address	Not managed for interchangeability purposes.
Note 1) Repeated plots, due to time synchronization constraints, must be re-updated. References are not translated from the original language.	

5.2.2 Message Definition

Must comply with:

- CEI EN 62056-21;
- ISO/IEC 13239;
- Green Book;
- Blue Book.

5.2.3 Awakening method

The optical port must be activated by pressing a button or by actuating a specific sequence of buttons.

Note Other methods are permitted but they are outside the scope of interchangeability.

5.3 Application Layer

5.3.1 Application Protocol

It must comply with:

- Green Book
- Blue Book
- UNI/TS 11291-11-2:2014.

5.3.2 DLMS/COSEM Client

5.3.2.1 Clients Supported by GdM and Repeaters

With reference to GdMs and Repeaters, the permitted local clients are indicated in table 3, taken from UNI/TS 11291-11-2:2014.

prospectus 3 **Clients allowed local access to GdMs/Repeaters**

DLMS Client	AA	Client ID	Association Type
Public Client	For relevant information see UNI/TS 11291-11-2:2014		Explicit
Client Management			Pre-established
Guarantor Authority			Pre-established
Client Maintenance			Explicit

Access to Client Management is required to enable Client Maintenance.

The Guarantor Authority requires access to change keys in the event of a change in the distributor's concession.

Access to the Maintenance Client is required to enable maintenance operations, including installation.

5.3.2.2 DC/GW Supported Clients

With reference to DC/GW, the permitted local clients are indicated in table 4, taken from UNI/TS 11291-11-2:2014.

prospectus 4 **Clients that are allowed local access to the DCs/GWs**

DLMS Client	AA	Client ID	Association Type
Public Client	For relevant information see UNI/TS 11291-11-2:2014		Explicit
Client Management			Explicit / Pre-established
Guarantor Authority			Pre-established
Client Maintenance			Explicit

Access to Client Management is required to enable Client Maintenance.

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

The Guarantor Authority requires access to change keys in the event of a change in the distributor's concession.

Access to the Maintenance Client is required to enable maintenance operations, including installation.

5.3.3 Safety

No security profiles are implemented at the physical or data link levels. Security is managed by the COSEM application layer, using HLS (High-Level Security) as specified in the Blue Book.

Pre-established associations do not need to go through the authentication process.

The maintainer uses the “user identification process” and the mechanism_id(5) authentication mechanism as described in point 9.2.3.4 Pass 3 of the Green Book. For further information see point 5.4.3 of UNI/TS 11291-11-2:2014.

Communication with the maintainer must be authenticated and encrypted. For further information, see section 5.4.3 of UNI/TS 11291-11-2:2014.

5.3.4 Factory mode

The optical port is often used in production for testing and programming purposes. This typically uses proprietary protocols for several reasons, including avoiding the need to hardcode all the proprietary objects in the COSEM DLMS (e.g., hardware configuration, calibration coefficients, etc.) for production purposes only.

For this purpose, a special mode of production is defined.

Referring to Table 6 of the Green Book, the Upper HDLC Address range from 0x0D-0x0F is reserved for this specific application mode. Frames with these addresses are exchanged with the proprietary application rather than the COSEM DLMS application.

The use of these addresses and the related application running on them are left to the manufacturers.

This manufacturing mode cannot benefit from the COSEM DLMS security model and could therefore represent a backdoor if left active in the field. For this reason, the manufacturer's application described in this chapter is only permitted during the manufacturing process and must be deactivated before the devices leave the factory.

6 WORKFLOWS

6.1

Pre-protected commands

The pre-protected controls are dedicated exclusively to maintenance operations.

A pre-protected command is a standard DLMS APDU, encrypted with the appropriate encryption key, entered into the device using the pre-established "Client Management" or "Guarantee Authority" association.

The command to activate the AA "Maintenance" has at least the following parameters to open a "maintenance window":

- departure date and time;

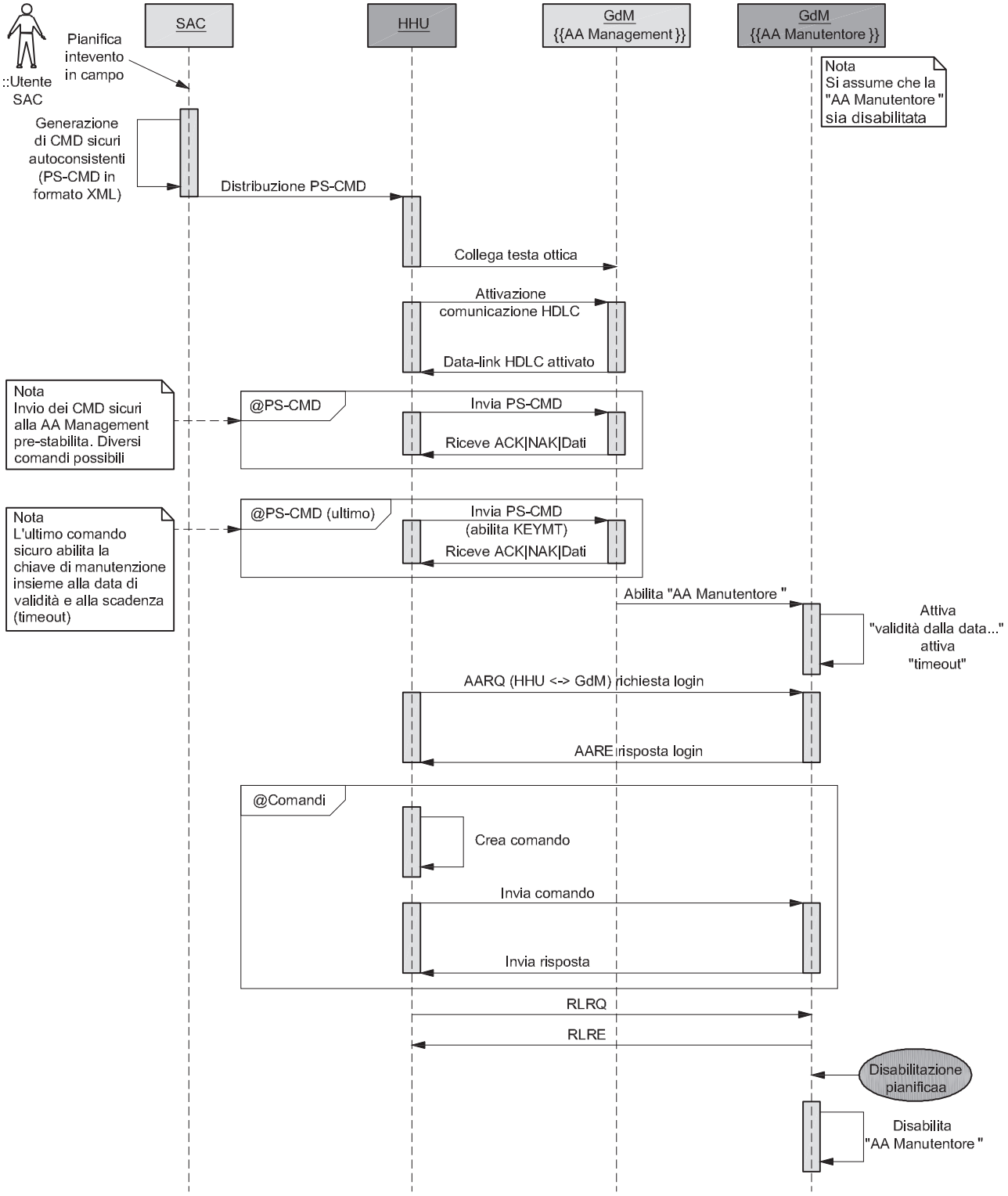
- validity period (for example expressed in minutes).

Figure 2 shows the interactions between different actors, before and during field operations.

A precondition is that the HHT has received pre-protected commands from the SAC for a given counter in a known format. This format is described in Appendix A.

© UNI

figure 2 **Sequence diagram showing interactions between different subjects, with pre-protected commands**



Other preconditions are as follows:

- a maintenance operation is required for a device (GdM / DC / GW / Repeater);

- correct pre-protected messages are created by the SAC, when the user requests it.

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

6.2 Safe Commands on the HHT

Pre-protected commands can only be decrypted and controlled by the owner of the key in question ("Client Management", i.e. KEYC).

There is no need to protect the commands on the HHT, since the technician cannot decode the message content in any way. There are no security keys, with the sole exception of the timed KEYMT, stored on the technician's HHT.

Furthermore, through proper use of the frame counter at the application level, rejecting frames with a counter that is less than or equal to the maximum value received by the device (and therefore resident there), the device itself is well defended against replay attacks.

The path that the SAC must take to process pre-protected messages is different and unrelated to that used to send commands via PP3/PM1. Therefore, appropriate recognition mechanisms must be implemented to prevent the local command from being rejected due to security criteria.

Note Since it is not known when a remote communication might take place, using the “frame counter”-based acceptance criterion could invalidate a pre-protected message, prepared before the communication, which would end up with a “frame counter” of a lower value than the one left after the remote communication, and vice versa: this would be interpreted as a “replay” attack.

Further protection is provided, as described in UNI/TS 11291-10, by rejecting any further communication for a programmable period of time after a certain number of attempts (default: three) to access with incorrect data or to repeat the same pre-protected message.

6.3 Installation procedure

6.3.1 GdM

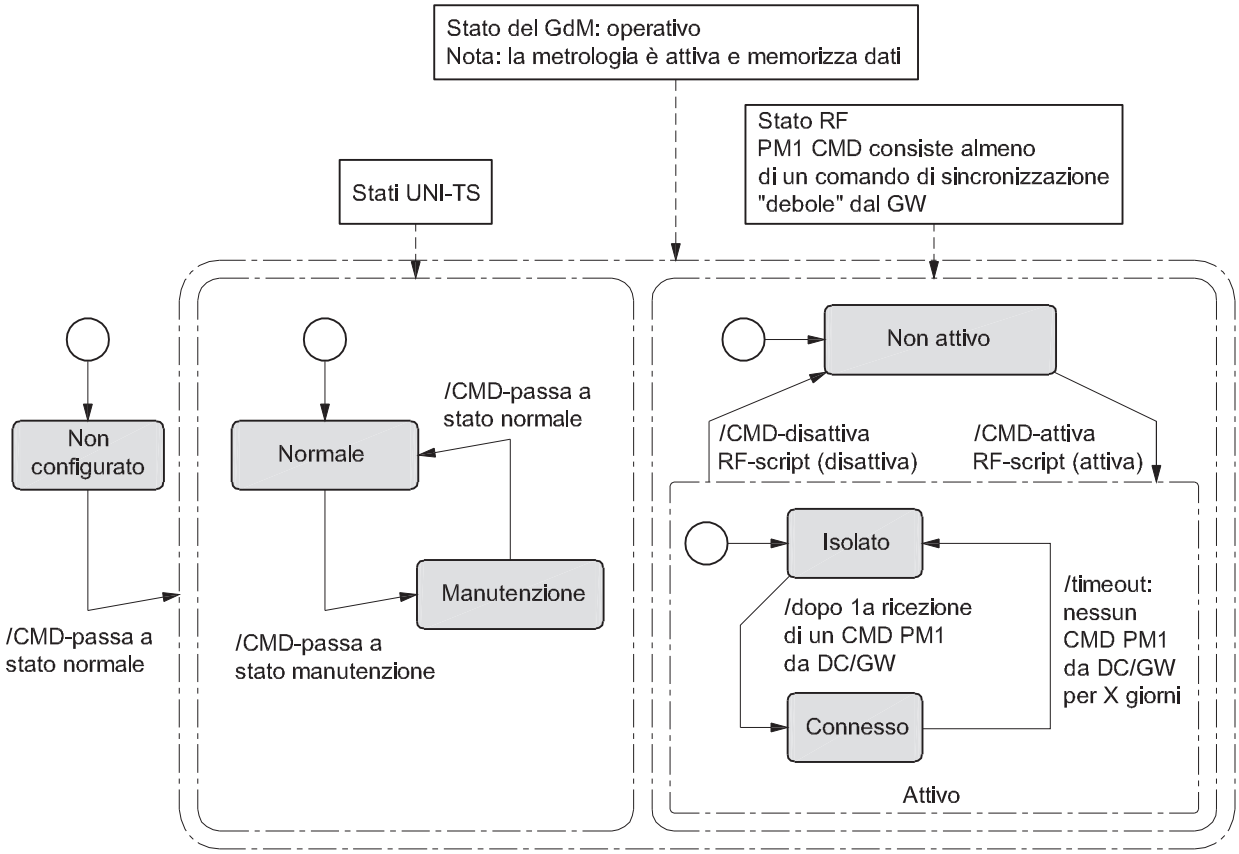
As defined in point 7.3 of UNI/TS 11291-6:2013, the device must have a “state”. The clock can be programmed (current date and time) without limitations in the “Not Configured” state.

Figure 3 shows the state diagram in accordance with the specific requirements of UNI/TS 11291 and the state of the RF management object (see UNI/TS 11291-11-2: 2014 for details).

Note In the meter, the two objects "UNI/TS status" and "RF status" can be managed independently.

© UNI

figure 3 **UNI/TS state diagram and RF state for PM1 devices (GdM and DC/GW/Repeaters)**



The items highlighted in Table 5 must be present in the device at the end of the installation.

prospectus 5 **Scheduled items that must be present at the end of the installation process**

Element	GdM	Repeater
Current date and time	Yes	Yes
Daylight Saving Time (DST) indication	Yes	Yes
Measuring Point ID (PDR)	Yes	
Parameters related to volume compensation (for example T_b, P_b)	Yes	
Billing Period Parameters	Yes	
End of day gas time	Yes	
Tariff schedule	Yes	
RF Configuration Parameters	Yes	Yes
UNI/TS State	Yes	Yes
RF State	Yes	Yes
Note These objects must be programmable through the Maintenance Client.		

6.3.2 Gateway/Concentrator

The same scheme in Figure 3 also applies to the GW/DC. For the data that must be present and configured at the end of the installation process (see Table 6), refer to UNI/TS 11291-11-2: 2014.

On network devices, the radio interface can only be configured and tested after the clock has been set correctly.

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

Note This requirement does not apply to the meter, where the radio can be configured and activated independently of the (local) clock state.

prospectus 6 **Scheduled items that must be present at the end of the installation process**

Element	Gateway	Concentrator
Current date and time	Yes	Yes
Daylight Saving Time (DST) indication	Yes	Yes
Parameters for configuring connectivity (GPRS, IP, TCP, ..., wake-up windows, ...)	Yes	Yes
UNI/TS State	Yes	Yes
Note These objects must be programmable through the Maintenance Client.		

6.3.3 Disabling the Client

The Maintenance Client must be disabled after the field process is complete.

Possible triggers include:

- Timeout after switching from "Not Configured" to "Normal" Installation;
- Timeout after switching from "Maintenance" to "Normal" Maintenance;
- Explicit command (e.g. session termination) or credential timeout to disable the client Revoke HLS credentials;
- Expiration of security credentials (temporary key);
- Session timeout.

In any case, at least the expiration of security credentials must disable the maintenance client.

6.4 Local actions

6.4.1 General information

Once the maintainer is granted access to the device and has logged in, he or she can perform any action that is permitted with respect to the client used to access the device.

Depending on the user's rights, the complete set of data models can be read or written, as described in UNI/TS 11291-11-2:2014, in addition to performing local clock synchronization.

UNI/TS 11291-11-2:2014 fully describes the access rights of the maintainer client.

6.4.2 Local actions that use only pre-protected commands

In some use cases, the HHU can be used as a simple repeater of commands/requests from the SAC. This allows you to avoid activating a local maintenance key for simple tasks like reading data or opening/closing valves. These tasks use pre-protected commands that connect to the Management client's pre-established association. These commands are protected with the corresponding KEYC.

The HHU is therefore not able to interpret the response from the device, consequently it will simply store the responses, in the same sequence as the sent commands, in an exchange file with the SAC.

6.5 Firmware Update

GdMs and repeaters must support local firmware updates. Firmware updates must be possible without a remote connection via PM1 or PP3.

The portable terminal behaves like a client using the Image Transfer Service (Chapter 4.4.6 of the Blue Book), with the exception that the initialization method and activation scheduling must not be allowed on the maintenance client and must be performed only under SAC authority.

To do this, the HHT carries pre-protected messages generated by the SAC. These messages are inserted into the network device (including the meter) and allow the HHT to perform all further operations required for the firmware update.

In order to perform the firmware update, an appropriate pre-protected command is provided by the SAC.

This pre-protected command must contain at least:

- activation of the firmware update;

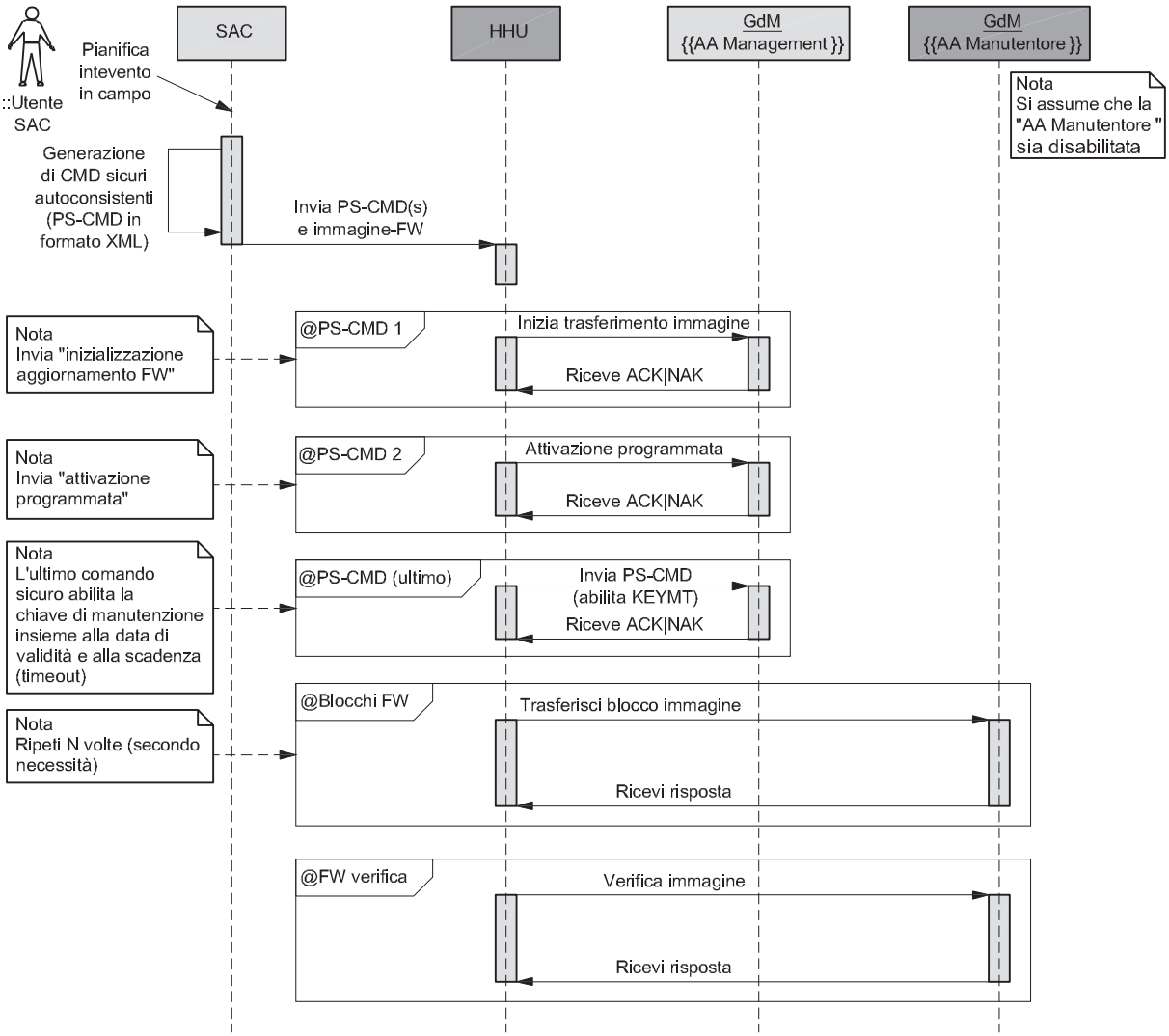
- activation time of the new firmware (see detailed information on activation planning in UNI/TS 11291-11-2: 2014).

Note The command with this "pre-programmed" activation can trigger activation at a predefined time in the future or immediately after the firmware image has been verified (i.e., "now"), to optionally allow field personnel to verify proper activation of the new image before leaving the installation site.

Pre-protected messages use the default client management, while all other commands are generated by the HHT using the maintenance client after a successful explicit association is triggered.

Pre-protected command operations must be executed before any other local operations to avoid exposing the process strategy. A typical workflow description is provided in Figure 4.

figure 4 **Local firmware update**

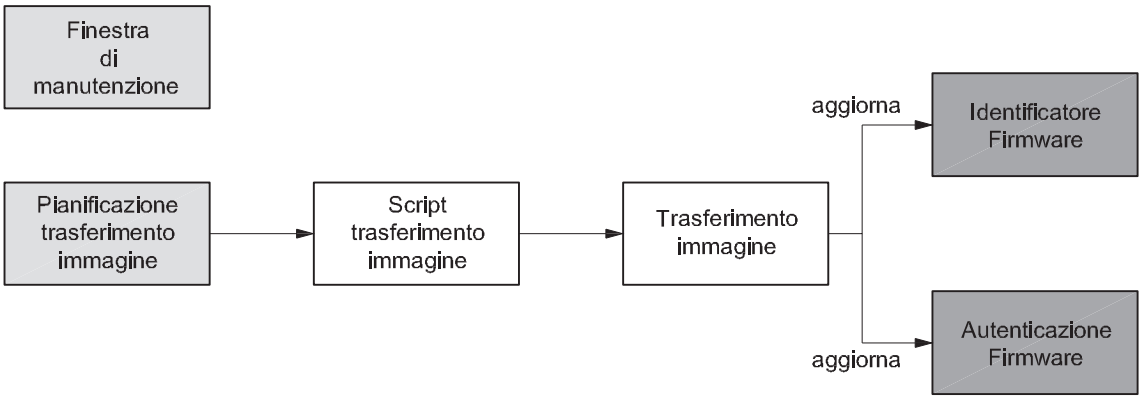


A set of objects must be available in the meter or repeater for this purpose, as exemplified in Figure 5.

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

figure 5 **Firmware update items**



These objects and their use are described in UNI/TS 11291-11-2: 2014.

Table 7 defines the steps for updating the firmware.

prospectus 7 **Firmware update steps**

Step	Client, how it works	Activity
1 Image block size	Maintenance Client This operation is not mandatory as it is specified in UNI/TS 11291-11-2: 2014.	Get image block size
2 Enable image transfer	Pre-protected command via client management Other operations can be performed via the maintenance client and the HHT	Action: Initiate image transfer Initiate_image_transfer Action: Set the activation scheduler (Image_transfer_scheduler [future instant | now]) Set the information needed for image verification (digital signature)
3 Transfer image block	Maintenance Client	Action transfer image block
4 Check completeness and send missing blocks	Maintenance Client Probably not necessary for local use	Not mandatory when confirmation is guaranteed for each transferred block.
5 Image Verification	Started by the GdM or network device when the block transfer is completed	The verification mode is not interchangeable. However, the image transfer states (*image_transfer_status*) must be handled correctly by the device
6 Check image	Mandatory within the scope of this document. The operator must have the response available upon download.	It uses the control parameters set at activation (digital signature). The methods of this control are not subject to interchangeability as they are free for each manufacturer.
7 Firmware activation	Must be provided in step 2 to allow a pre-protected command before any other action	Action: Activate image

6.6 Management of manufacturer-specific elements

Manufacturer-specific object and interface class definitions shall comply with Blue Book 4.1.4.

Manufacturer-specific objects must have specific OBIS codes (see Blue Book 5.2.8 - Manufacturer-specific codes), in order to have a univocal interpretation of the object definition.

When a list of objects is requested from a client (“read object list”), all visible objects, including manufacturer-specific objects, are visible, but the data model definition of those objects is left to the manufacturer's choice.

The behavior of the counter after writing a new value to a specific object/attribute is specified by the manufacturer.

APPENDIX (regulations) EXCHANGE FORMAT FOR PRE-PROTECTED COMMANDS AND RESPONSES

This appendix defines the file format that must be used to export the pre-protected commands generated by the SAC and import the generated responses. These commands are transported via XML files.

The content of the above XML file, from SAC to HHU is as follows: 1)

Unique device identifier.

a) It is the unique device number for devices using the protocol, exposed by the unique logical device name composed as defined in point 5.3.2 of UNI/TS 11291-11-2:2014:

i) manufacturer

ii) Type of device

iii) Device version

iv) serial number

2) ASCII-HEX representation of the temporary key (KEYMT) to be delivered to the maintainer.

3) ASCII representation of the password to be used on the device by the maintainer.

4) List of pre-protected commands

a) At least one pre-protected command must be

i) Command to change KEYMT

ii) Command to activate the AA starting from a specific date and time (validity timestamp).

iii) Command to program the AA to be deactivated after a given timeout (timeout validity).

iv) In case the Maintainer could be dynamically granted rights on that GdM/DC/GW/Repeater, the layout of the rights must also be provided.

b) All above commands are encrypted with KEYC of GdM, DC/GW, repeater.

c) Validity timestamp, validity timeout and rights must be displayed next to the pre-protected command.

d) Each pre-protected command must be numbered in the XML file.

The representation in table form is as follows.

prospectus A.1

Information	Example	Type	Domain
Unique device identifier	YWZ-X-00-00-123456789		
KEYMT	AAC55F0CAC5C51894D459573ED033E3A	ASCII-HEX	[0-9A-F-f]{32,32}
Password	ABCDEFGH	ASCII	
Time stamp validity	2013-03-31T11:23:00+02	ISO 8601 Time Stamping	[YYYY]-[MM]-[DD]T[hh]:[mm]:[ss] ± [hh]
Timeout Duration	PT03H	ISO 8601 Duration	PT[hh]H
Rights	0A34EF12	ASCII-HEX representation of a 32-bit bitmask	
Message Number	1		
Pre-protected message	1234567890...AABBCCDDEEFF	ASCII-HEX	[0-9A-F-f]+
...			
Message Number	23		
Pre-protected message	AABBCCDDEEFF...1234567890	ASCII-HEX	[0-9A-F-f]+

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

In the opposite direction, the format whose representation in table form is as follows must be used:

prospectus A.2

Information	Example	Type	Domain
Unique device identifier	YWZ-X-00-00-123456789	As actually read by the device	
Message Number	1		
Timekeeper	2013-03-31T11:23:00+02	ISO 8601 timestamp	[YYYY]-[MM]-[DD]T[hh]:[mm]:[ss] ±[hh]
Answer (safe)	1234567890...AABBCCDDEEFF	ASCII-HEX	[0-9A-F-f]+
...	...	...	...
Message Number	23		
Timekeeper	2013-03-31T11:23:00+02	ISO 8601 timestamp	[YYYY]-[MM]-[DD]T[hh]:[mm]:[ss] ±[hh]
Answer (safe)	AABBCCDDEEFF...1234567890	ASCII-HEX	[0-9A-F-f]+

© UNI

APPENDIX B HDLC FRAME STRUCTURE

(regulations)

B.1 HDLC textures

B.1.1 General information

HDLC frames are composed of:

- a Flag;
- a Header;
- an address field;
- a Command field;
- an HCS (header check sequence);
- the information field;
- the FCS (frame check sequence field).

Not all frames have an information field. If there's no information field, then there's no FCS field either.

B.1.2 The Flag

It has value 0x7E.

B.1.3 The Header

It is 2 bytes in size. A TYPE 3 header is used. The header consists of three fields:

- the type with bits from 15 to 13. Its value is 0b101;
- the segmentation field: bit 12. It is FALSE if the frame is not segmented, otherwise it is TRUE;
- length: from bit 12 to bit 0, the chd leads to a maximum value of 4,096 bytes. DLMS/ Cosem limits this value to 2,030 bytes, but in this specification this value is limited to 1,024. The length takes into account the header up to the FCS. Flags are excluded.

B.1.4 The address field

Two addresses must be defined: the source address and the destination address. These are the client address and the server address.

- the client address fits on a single byte. This is the client ID. Its values range from 0 to 0x7E;
- the server address is made up of two fields: they can be 1 or 2 bytes each:

  - the physical address,
  - the address of the logical device.

B.1.5 The command field

It consists of one byte. The most common commands are shown in Table B.1.

© UNI

table B.1

Command	Value	Primary station (client)	Answer	Value	Secondary Station (server)
SNRM	100P0011	Set Normal mode	UA	011F0011	Unnumbered acknowledge
DISC	010P0011	Disconnect			
THE	RRRP/F|SSS0	Information	THE	RRRP/F|SSS0	Information
RR	RRRP/F0001	Receiver Ready	RR	RRRP/F0001	Receiver Ready
FRMR	100F0111	Frame reject			

B.1.6 HCS

It allows you to verify the integrity of the message from the header to the command. The calculation is the same as the FCS.

B.1.7 Information

The information field contains the useful data to be transmitted. This field has variable dimensions: for the purposes of this document, a maximum length of 1,024 bytes is used.

B.1.8 FCS

FCS allows for verification of the integrity of the information field. See point 8.5 of the Green Book.

B.1.9 Flag

Another flag field ends the frame.

B.1.10 Example

Below is an example highlighting the plot fields:

7EA04502230310FB7FE6E6006036A1090607608574050801018A0207808B07608574
05080201AC0A8008313233343536373 8BE10040E01000000065F1F04001C1E3D0000
2CCB7E

- First the Flag (7E);
- Next comes the Header, which in this case informs of the existence of 0x45=60 bytes in the frame;
- Below is the address field: the destination address is that of the server composed of two parts: logical device (value 1) and physical device (value 0x11=17);
- The command (10) which in this case is: information;
- HCS which is 0xFB7F;
- The information field;
- FCS relating to the information field;
- Closes the message with the Flag (7E).

B.1.11 Communication sequence

Below is a communication sequence that starts with a connection made with the SNRM command, 3 sequences of information exchange and, at the end, the disconnection.

7EA01F02230393A2F08180120501FF0601FF07040000000108040000000763797E

7EA0210300020023737B7F818012050180060180070400000001080400000001533B7E

7EA04502230310FB7FE6E6006036A1090607608574050801018A0207808B076085
7405080201AC0A8008313233343536373 8BE10040E01000000065F1F04001C1E3D00
002CCB7E

© UNI

7EA047030002002330ACF3E6E7006136A1090607608574050801 01A203020100A305A1030201008802078089076085740508020 1BE10040E0800065F1F0400001E1D04F8000717ED7E

7EA0080223033126F27E

7EA00A0300020023316C0B7E

7EA01A022303327562E6E600C001C100010101000200FF02009E247E

7EA0230300020023524B78E6E700C401C100090E4954453432314732010701100028 7DCA7E

7EA0080223035120917E

7EA00A0300020023516A687E

7EA0080223035332B27E

7EA00A0300020023737A6A7E

B.1.12 Explanation

SNRM

7EA01F02230393A2F08180120501FF0601FF0704000000010804000000076379 7E UA

7EA0210300020023737B7F818012050180060180070400000001080400000001533B7E

Information

7EA04502230310FB7FE6E6006036A1090607608574050801018A0207808B0760857405080201AC0A8008313233343536373 8BE10040E01000000065F1F04001C1E3D00002CCB7E

Information

7EA047030002002330ACF3E6E7006136A1090607608574050801 01A203020100A305A1030201008802078089076085740508020 1BE10040E0800065F1F0400001E1D04F8000717ED7E

Receiver Ready

7EA0080223033126F27E

Receiver Ready

7EA00A0300020023316C0B7E

Information

7EA01A022303327562E6E600C001C100010101000200FF02009E247E

Information

7EA0230300020023524B78E6E700C401C100090E4954453432314732010701100028 7DCA7E

Receiver Ready

7EA0080223035120917E

Receiver Ready

7EA00A0300020023516A687E

Disconnection

7EA0080223035332B27E

UA

7EA00A0300020023737A6A7E

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

BIBLIOGRAPHY

UNI/TS 11291-8 Gas measurement systems - Hourly gas measurement devices - Part 8: Protocols for remote management of metering units for the distribution network

UNI/TS 11291-11-1 Gas measurement systems - Hourly gas measurement devices - Part 11-1: General

IEC EN 62056-42 Electricity metering - Data exchange for meter reading, load tariff control - Part 42: Physical layer services and procedures for asynchronous and connection-oriented data exchange

IEC EN 62056-46 Electricity metering - Data exchange for meter reading, load tariff control - Part 46: Data link layer using the HDLC protocol

IEC EN 62056-53 Electricity metering - Data exchange for meter reading, load tariff control - Part 53: COSEM application layer

AEEG Resolution ARG/gas155/08 “Guidelines for the commissioning of gas measurement units, characterised by minimum functional requirements and with remote reading and remote management functions, for the delivery points of natural gas distribution networks” Annex A

AEEG Resolution 631/2013/R/GAS Amendments and additions to the obligations for commissioning gas smart meters

DLMS User Association 1000-2 - Green Book - DLMS/COSEM Architecture and Protocols - 7th ed. and related Amendment 3

DLMS User Association 1000-1 - Blue Book COSEM - Identification System and Interface Classes - 11th ed.

© UNI

ELSEWEDY ELECTROMETER EGYPT
UNIstore - 790840

UNI
Italian National Unification Body
Via Sannio, 2
20137 Milan, Italy

Reproduction prohibited - Law 22 April 1941 No. 633 and subsequent updates.